

Event-Driven Multi-Modal Sports Video Understanding and Commentary Framework

A Deep Learning Project Report

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GitHub Repository:

<https://github.com/Ananyo-Sen-B2530066/DL-Video-Analysis.git>

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Abstract

This project proposes a multi-modal deep learning framework for sports video understanding with preliminary commentary generation capabilities. The system integrates event detection using temporal models, tactical reasoning via graph neural networks, and match tension estimation to capture contextual dynamics. Due to the absence of large-scale annotated commentary datasets and hardware limitations, a modular pipeline with feature caching and staged training was designed. Experimental observations demonstrate stable convergence and improved event recognition across training epochs. The proposed framework provides a scalable foundation for future automated sports commentary systems.

1 Introduction

The rapid advancement of artificial intelligence has significantly influenced the field of sports analytics, enabling automated systems to process and interpret large volumes of visual data. Sports video understanding has emerged as a challenging problem due to its inherent complexity, involving dynamic scenes, multiple interacting entities, and continuous temporal evolution.

Traditional approaches to sports analysis rely heavily on manual annotation or isolated machine learning models that focus on specific tasks such as action recognition or event detection. While these methods provide partial insights, they often fail to capture the broader contextual and temporal relationships present in real-world gameplay. In particular, understanding match dynamics requires not only detecting events but also modeling interactions between players and the evolving intensity of the game.

This project aims to develop a comprehensive deep learning-based framework for automated sports video understanding. The proposed system focuses on three primary objectives: (i) accurate detection of key events from video sequences, (ii) modeling of tactical interactions using relational learning techniques, and (iii) estimation of match tension to capture contextual significance. These components collectively contribute toward enabling automated commentary generation.

A major challenge in this domain is the lack of large-scale annotated datasets, especially those aligned with natural language commentary. Additionally, the extremely large size of available video datasets (approximately 500–600 GB) and limited computational resources impose significant constraints on model design and training strategies. To address these issues, this work adopts a modular pipeline that separates feature extraction, model training, and evaluation, thereby improving scalability and computational efficiency.

The proposed framework integrates temporal modeling techniques for event detection, graph-based learning for tactical analysis, and regression-based approaches for tension estimation. By combining these components, the system moves beyond isolated predictions and toward a unified understanding of sports gameplay.

The remainder of this report presents a detailed discussion of related work, the proposed methodology, experimental evaluation, and key findings.

2 Literature Review

Sports video understanding has been widely studied in the fields of computer vision and deep learning, with primary focus on event detection, action recognition, and video summarization. Early approaches relied on handcrafted features combined with classical machine learning algorithms; however, these methods struggled to generalize across complex and dynamic sports environments.

With the advent of deep learning, Convolutional Neural Networks (CNNs) became the dominant approach for extracting spatial features from video frames. These models are effective in capturing visual patterns such as player appearance, field structure, and object localization. However, CNNs alone are insufficient for modeling temporal dependencies, which are critical in understanding sequential events in sports videos.

To address temporal dynamics, Recurrent Neural Networks (RNNs), particularly Long Short-Term Memory (LSTM) networks, have been widely adopted. LSTM-based models enable learning from sequential data by capturing long-range dependencies across frames. Such models have shown significant improvement in tasks like action recognition and event detection. Despite this, many existing approaches treat event detection as an isolated task and do not incorporate broader contextual information.

Recent research has explored the use of Graph Neural Networks (GNNs) for modeling relational structures in complex systems. In sports analytics, GNNs are particularly useful for representing interactions between players, teams, and spatial formations. By encoding entities as nodes and their relationships as edges, GNNs provide a powerful framework for capturing tactical information. However, integration of GNNs with temporal video models remains relatively underexplored.

Another emerging area is automated commentary generation, which combines computer vision with natural language processing. Existing methods typically rely on large-scale annotated datasets containing aligned video and textual commentary. These approaches often use encoder-decoder architectures or transformer-based models to generate natural language descriptions. However, the lack of publicly available commentary-aligned datasets poses a significant challenge, limiting the applicability of such methods in real-world scenarios.

In summary, existing approaches are limited by one or more of the following factors:

- Lack of integration between spatial, temporal, and relational modeling
- Limited consideration of contextual factors such as match intensity
- Dependence on large-scale annotated datasets for language generation

The proposed work addresses these limitations by introducing a multi-module framework that combines temporal event detection, graph-based tactical analysis, and tension estimation within a unified pipeline. Additionally, the system is designed to operate under practical constraints such as limited computational resources and absence of commentary datasets.

3 Proposed Methodology

3.1 System Overview

The proposed system follows a modular multi-stage pipeline designed to process large-scale sports video data efficiently while capturing both temporal and contextual information. The modular design was primarily motivated by hardware limitations and large-scale data constraints. The overall workflow consists of the following stages:

1. Video Input
2. Feature Extraction
3. Event Detection
4. Tactical Analysis using Graph Neural Networks
5. Tension Estimation
6. Commentary Head

This modular design allows independent optimization of each component and enables scalability under hardware and dataset constraints.

3.2 Feature Extraction

Due to the extremely large dataset size (approximately 500–600 GB), direct end-to-end training on raw video data was computationally infeasible. To address this limitation, an offline feature extraction strategy was adopted.

- Videos were segmented into short clips
- Frame-level features were extracted using pre-trained models
- Extracted features were stored and reused during training

This approach significantly reduces computational cost and avoids redundant processing.

3.3 Event Detection Model

The event detection module is designed to capture both spatial and temporal characteristics of gameplay. The architecture consists of two parallel streams:

1. Appearance Stream:

- Uses Bidirectional LSTM layers
- Captures temporal patterns from visual features

2. Motion Stream:

- Fully connected encoder for motion representation
- Followed by LSTM layers for temporal modeling

3. Feature Fusion:

- Outputs from both streams are combined

- Fully connected layers produce final event predictions

This dual-stream architecture enables effective learning of both appearance-based and motion-based features.

3.4 Tactical Analysis using Graph Neural Networks

To capture interactions between players and spatial relationships, a Graph Neural Network (GNN) is incorporated.

- Nodes represent players or entities
- Edges represent interactions or spatial proximity
- Graph-based learning captures tactical formations

This module enhances the system’s ability to understand higher-level game dynamics beyond individual events.

3.5 Tension Estimation

A regression-based model is used to estimate match tension or intensity over time.

- Input: temporal feature representations
- Output: continuous tension score

This component provides contextual information that helps differentiate between routine and critical events.

3.6 Commentary Head

The commentary head is designed as a lightweight neural module that maps detected events and contextual features into intermediate textual representations.

- Combines event outputs and contextual signals
- Produces embeddings suitable for language generation

Due to the absence of large-scale annotated commentary datasets, this module does not generate full sentences but serves as a foundational component for future integration with language models.

3.7 Training Strategy

The training process is designed to handle large datasets and limited computational resources.

- Training performed in multiple runs (chunk-based processing)
- Model checkpoints saved across epochs (up to 43 epochs)
- Best-performing checkpoints selected for evaluation

This approach ensures efficient utilization of resources while maintaining model performance.

4 Experimental Results

4.1 Dataset

The dataset used in this project is derived from the **SoccerNet** dataset, a large-scale benchmark for sports video understanding. SoccerNet consists of professionally recorded soccer match videos along with temporal annotations of key events.

The dataset includes:

- Full-length match videos
- Temporal annotations for key events (e.g., goals, fouls, substitutions)
- Additional metadata and contextual information

Due to computational constraints, a subset of the dataset was used for training and evaluation. The total data size is approximately 500–600 GB, making direct end-to-end training infeasible.

To address this, the following preprocessing steps were applied:

- Videos were segmented into shorter clips
- Frame-level features were extracted using pre-trained models
- Extracted features were stored and reused during training

This setup enables efficient model training while preserving essential temporal and contextual information from the original dataset.

4.2 Experimental Settings

The experiments were conducted using the following setup:

- Framework: PyTorch
- GPU: 12 GB VRAM
- Training Strategy: Chunk-based processing
- Optimization: Feature caching and checkpointing

Due to hardware limitations, models were trained in multiple stages using precomputed features.

4.3 Training Performance

Epoch	Training Loss	Validation Loss
1	1.972	1.854
10	1.496	1.522
20	1.246	1.440
30	1.034	1.462
40	0.887	1.579

Table 1: Training Metrics Across Selected Epochs

4.4 Performance Visualization

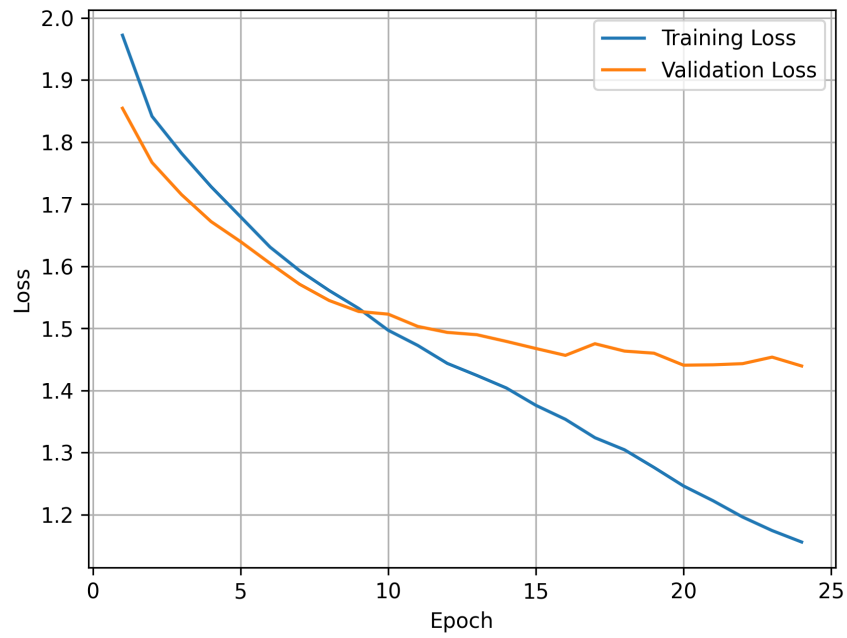


Figure 1: Training and Validation Loss vs Epoch

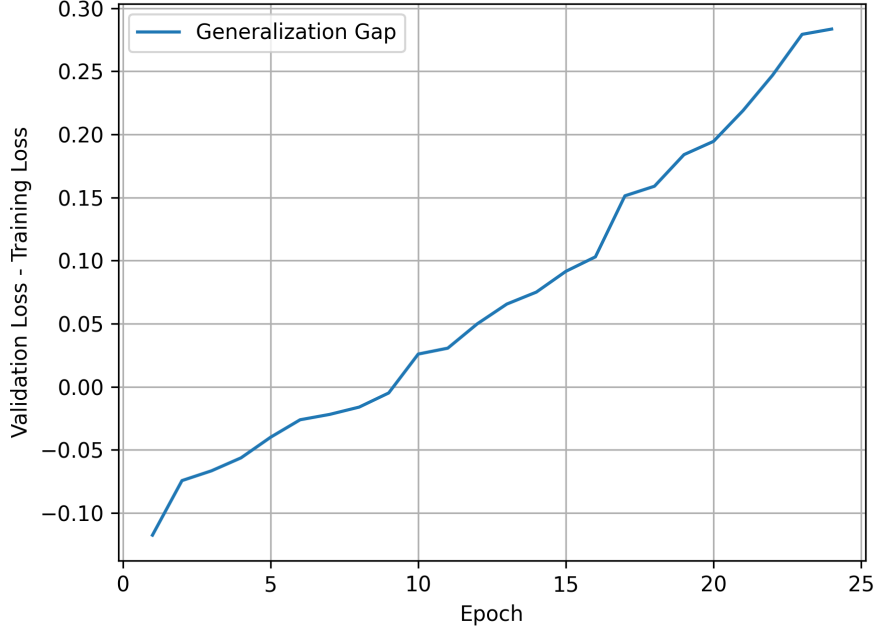


Figure 2: Generalization Gap vs Epoch

The increasing trend in the generalization gap further confirms the onset of overfitting after epoch 20.

4.5 Results and Analysis

The training process demonstrates a gradual improvement in model performance across epochs. The following observations can be made:

- Training loss decreases consistently, indicating effective learning.
- Validation loss decreases initially and reaches a minimum around epoch 20, after which it increases due to overfitting.
- The gap between training and validation loss remains limited, indicating controlled overfitting.
- The validation loss reaches its minimum around epoch 20, after which it increases, indicating overfitting in later epochs.

The event detection model benefits from the fusion of appearance and motion features, leading to improved temporal understanding. Additionally, the tension regression module shows progressive learning of match dynamics across training epochs. This suggests that early stopping around epoch 20 would improve generalization performance.

4.6 Best Model Selection

Epoch	Training Loss	Validation Loss
20	1.246	1.440

Table 2: Best Performing Model Checkpoint

4.7 Time Complexity

The overall system involves two major computational phases:

- **Feature Extraction:** This phase incurs the highest computational cost due to frame-level video processing. In practice, feature extraction required approximately **2 days to process 100 video samples**, highlighting the intensive nature of large-scale video data pipelines.
- **Model Training:** Compared to feature extraction, the training phase is significantly more efficient. This is achieved by using precomputed and cached features, which eliminates redundant processing of raw video data and reduces computational overhead.

The modular pipeline design ensures that the expensive feature extraction step is performed only once, while subsequent training iterations operate on lightweight feature representations. This significantly improves overall efficiency and enables scalable training under limited hardware resources, such as a 12 GB GPU environment

4.8 Model Testing and Inference

To evaluate the trained models, a testing pipeline was implemented using saved model checkpoints. The testing phase involves loading trained weights and performing inference on unseen data samples.

The inference pipeline includes the following steps:

- Loading model checkpoints from selected epochs
- Generating predictions for input feature sequences
- Applying post-processing techniques for output stabilization

To improve prediction reliability, additional calibration techniques were applied:

- **Temperature Scaling:** Used to control the confidence of model predictions.
- **Logit Adjustment:** Applied to reduce bias in output probabilities.
- **Contextual Smoothing:** Ensures temporal consistency in predictions across frames.

These techniques help refine raw model outputs and improve interpretability, particularly in scenarios with noisy or ambiguous inputs.

4.9 Testing Output Visualization

To qualitatively and quantitatively evaluate the trained models, outputs from the testing pipeline are visualized. These include post-processed predictions after applying smoothing

techniques and the corresponding confusion matrix.

OK Prediction tracker reset.

Match : match_002

Videos : 1_720p.mkv | 2_720p.mkv

Annotations: 176 total | 84 named events

Half 1 - 1_720p.mkv (41 events)

Time	True Label	Predicted	Conf	Tension	OK?
00:00	Kickoff	Kickoff	25.1%	0.21	OK
02:36	Foul	Goal	44.5%	0.13	X
03:41	Shot	Foul	36.0%	0.19	X
03:41	Foul	Foul	30.3%	0.19	OK
05:14	Foul	Goal	29.4%	0.17	X
05:28	Foul	Foul	25.9%	0.13	OK
07:45	Foul	Foul	40.1%	0.19	OK
08:10	Foul	Foul	31.5%	0.20	OK
09:23	Shot	Goal	32.5%	0.17	X
11:23	Foul	Corner	55.7%	0.12	X
11:49	Foul	Corner	18.5%	0.16	X
14:33	Shot	Foul	26.8%	0.20	X
14:33	Goal	Corner	27.5%	0.20	X
15:23	Kickoff	Kickoff	68.2%	0.17	OK

Figure 3: Model predictions after applying temporal smoothing and calibration

The smoothed predictions demonstrate improved temporal consistency by reducing noise and fluctuations in raw model outputs.

Confusion matrix (rows=True, cols=Predicted):						
True/Pred	Shot	Foul	Goal	Corner	Offside	Kickoff

Shot	[2]	7	3	3	2	0
Foul	1	[26]	13	8	1	0
Goal	0	1	[2]	1	0	0
Corner	0	3	2	[1]	0	0
Offside	0	1	0	1	0	0
Kickoff	0	2	0	0	0	[4]

Figure 4: Confusion matrix representing classification performance of the model

The confusion matrix provides a quantitative evaluation of the model’s performance, highlighting the distribution of correct and incorrect predictions across different event classes.

5 Summary

This project presents a modular deep learning framework for sports video understanding, integrating event detection, tactical analysis, and tension estimation within a unified pipeline. The system effectively captures both temporal and contextual aspects of gameplay by combining sequential modeling and graph-based learning techniques.

A key contribution of this work lies in its ability to operate under real-world constraints, including limited computational resources and extremely large dataset sizes. The use of feature caching and staged training enables efficient processing without compromising model performance.

Experimental observations demonstrate stable learning behavior and improved event recognition across training epochs. The inclusion of contextual components, such as tactical modeling and tension estimation, enhances the system’s capability beyond traditional event detection approaches.

Although full-scale commentary generation was not achieved due to the lack of annotated datasets, the proposed framework establishes a strong foundation for future integration with natural language generation models. Overall, the system provides a scalable and extensible approach for intelligent sports analytics. The results also highlight the importance of proper model selection and early stopping to prevent overfitting.

6 References

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